Comparison of Prevalences across Groups

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Contents

1 Generating the Data

```
library(ggplot2)
```

We will use simulate data on cancer occurrence across smoker and non-smokers.

```
set.seed(123)

# Parameters
n_total <- 300
prob_smoker <- 0.50           # Overall smoking rate
prob_cancer_given_smoker <- 0.67      # P(Cancer|Smoker) = 100/150
prob_cancer_given_nonsmoker <- 0.133  # P(Cancer|Non-smoker) = 20/150

# Sample all variables using sample()
study_data <- data.frame(
    ID = sample(10000:99999, n_total, replace = FALSE),
  Smoking = sample(c("Smoking", "Non smoking"),    # Sample smoking status
                n_total,
                replace = TRUE,
                prob = c(prob_smoker, 1 - prob_smoker))
)

# Sample cancer status based on smoking (vectorized approach)
study_data$Cancer <- ifelse(
  study_data$Smoking == "Smoking",
  sample(c("Cancer", "No cancer"),
         sum(study_data$Smoking == "Smoking"),
         replace = TRUE,
         prob = c(prob_cancer_given_smoker, 1 - prob_cancer_given_smoker)),
  sample(c("Cancer", "No cancer"),
         sum(study_data$Smoking == "Non smoking"),
```

```r
         replace = TRUE,
         prob = c(prob_cancer_given_nonsmoker, 1 - prob_cancer_given_nonsmoker))
)

# 2. Explore the data
head(study_data)
```

2 Visualizing and Summarizing Data

2.1 Contingency Table

Generate a contingency table of cancer by smoking statuses.

```r
table(study_data$Smoking, study_data$Cancer)
```

```
##
##                Cancer No cancer
##   Non smoking      16       130
##   Smoking         105        49
```

```r
prop.table(table(study_data$Smoking, study_data$Cancer))
```

```
##
##                    Cancer  No cancer
##   Non smoking 0.05333333 0.43333333
##   Smoking     0.35000000 0.16333333
```

```r
addmargins(table(study_data$Smoking, study_data$Cancer))
```

```
##
##                Cancer No cancer Sum
##   Non smoking      16       130 146
##   Smoking         105        49 154
##   Sum             121       179 300
```

2.2 Probabilities

```r
# Smokers
total = sum(study_data$Smoking == "Smoking")
cancer = sum(study_data$Smoking == "Smoking" & study_data$Cancer == "Cancer")
prop.test(cancer, total, conf.level = 0.95)

# Non-smokers
total = sum(study_data$Smoking == "Non smoking")
cancer = sum(study_data$Smoking == "Non smoking" & study_data$Cancer == "Cancer")
prop.test(cancer, total, conf.level = 0.95)
```

The probability (95% confidence interval) of having a cancer is 0.68 [0.61; 0.75] in smokers, and 0.11 [0.07; 0.17] in males.

2.3 Interpretation

We expect an association between smoking status and cancer status as the confidence intervals for the probability of having a cancer across smoking statuses do not overlap.

3 Hypothesis Testing

3.1 Biological question

We aim to study the association between smoking status and cancer status. The null hypothesis is that there is no association.

3.2 Sample and Assumptions

Both variables are recorded as categorical variables.

```
# Check expected sample sizes
chisq.test(study_data$Smoking, study_data$Cancer)$expected
```

The expected samples sizes in each group under the null hypothesis are > 5, so a chi2 test is appropriate.

3.3 Inference with chi2 Test

```
# Run test
chisq.test(study_data$Smoking, study_data$Cancer)
```

3.4 Check and Explanation

A chi2 test rejected the null hypothesis (chi2 $= 99.612$, $p < 0.05$), i.e., we detected an association between cancer occurrence and smoking status. The probability (95% confidence interval) of having a cancer is 0.68 [0.61; 0.75] in smokers, and 0.11 [0.07; 0.17] in males.

4 Alternative Tests

4.1 Fisher Exact Test

Appropriate for smaller sample sizes

```
# Run test
fisher.test(study_data$Smoking, study_data$Cancer)
```

Here, we reach the same conclusion as above.

4.2 `prop.test()`

Uses other existing tests (e.g. chi2)

```
# Run test
# More difficult to use correctly, the successes (Yes) should be
# 1st in the contingency table
prop.test(table(study_data$Smoking, study_data$Cancer)[, c("Cancer", "No cancer")])
```

Here, it used a chi2, and we reach the same conclusion as above.